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S0FTWARE DEVELOPMENT CLUB

LEVEL 1

REPORT

ON

"STUDENT PERFORMANCE TRACKER"

Submitted partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Submitted By

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MVJ COLLEGE OF ENGINEERING

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MVJ COOLEGE OF ENGINEERING, BENGALURU – 560067
(Autonomous Institute Affiliated To VTU, Belagavi)
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the SDC Project work survey titled "**STUDENT PERFORMANCE TRACKER**" is carried out by us who are Bonafide students of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of Bachelor of Engineering in Computer Science and Engineering of the Visvesvaraya Technological University, Belagavi during the year 2025 - 2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report deposited in the departmental library. The project report has been approved as satisfies the academic requirements in respect of project work prescribed by the institution for the said Degree.

Signature of Guide	**Signature of HOD**
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DECLARATION

We, SHREYA R [1MJ24CS164], DEEPIKA N [1MJ24EA011], SHREYAS C [1MJ24CS218], THANUSHA C [1MJ24CS185] students of Third Semester B.E., Department Of Computer Science And Engineering, Department of Electronics and Communication Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the AEC Level I Literature survey ttitled '**STUDENT PERFORMANCE TRACKER'** has been carried out by us and submitted in partial fulfilment for the award of Degree of **DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING, DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING** during the year 2025-2026.

Further we declare that the content of the survey has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare tha tany Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

Place:Bengaluru

Date:

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ABSTRACT

The Student Performance Tracker (SPT) is a comprehensive software-based system developed to monitor, analyze, and enhance student academic performance through the systematic collection and evaluation of academic, behavioral, and attendance-related data. In modern educational environments, the increasing volume of student data and the demand for personalized academic support have highlighted the limitations of traditional performance monitoring methods. The proposed system addresses these challenges by providing an integrated and automated platform that enables efficient tracking, analysis, and visualization of student performance metrics.

The primary objective of this pre-report is to present a detailed overview of the proposed Student Performance Tracker, including its background, problem definition, theoretical foundation, environmental analysis, system design, and requirement analysis. This report serves as a foundational document for the complete project report and aims to establish a clear understanding of the problem domain, existing research studies, and the technical approach adopted for system development. By reviewing relevant literature and analyzing current practices, the report identifies gaps in existing student performance monitoring systems and justifies the need for a more robust and user-friendly solution.

The Student Performance Tracker is designed to assist educational institutions, faculty members, administrators, and students by providing real-time access to performance data, historical trends, and analytical insights. The system supports informed decision-making by enabling educators to identify academically at-risk students at an early stage and implement timely interventions. For students, the system promotes self-assessment and accountability by offering transparent access to their academic progress and performance patterns.

The proposed system incorporates modern data management techniques, secure databases, and analytical tools to ensure accuracy, reliability, and scalability. It emphasizes usability, data security, and compliance with educational data protection standards. This pre-report outlines the conceptual, functional, and structural framework of the project across five major chapters, forming the basis for detailed system implementation and evaluation in subsequent phases. The Student Performance Tracker ultimately aims to contribute to improved academic outcomes, enhanced teaching effectiveness, and data-driven educational management.

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CHAPTER 1
INTRODUCTION

1.Introduction

In today's technology-driven educational environment, effective monitoring of student performance has become essential for improving academic outcomes and institutional efficiency. Educational institutions are increasingly shifting from traditional manual methods to digital systems to manage academic data more accurately and efficiently. A Student Performance Tracker provides a structured and centralized approach to recording, analyzing, and evaluating student performance by integrating academic results, attendance, and related metrics. Such a system supports educators and administrators in making informed decisions, enables timely academic interventions, and empowers students to actively track and improve their learning progress.

1.1 Background of the Study

In recent years, educational institutions have increasingly adopted digital technologies to manage academic and administrative activities. The rapid growth of information technology, coupled with the expansion of student populations, has significantly transformed traditional teaching and learning environments. Conventional methods of tracking student performance—such as manual record keeping, paper-based files, or basic spreadsheet applications—are often inefficient, time-consuming, and prone to human error. These methods also lack the capability to provide real-time insights and comprehensive performance analysis.

As class sizes continue to grow and learning styles become more diverse, educators face increasing challenges in monitoring individual student progress effectively. Students differ in their academic abilities, learning pace, and engagement levels, making it essential for institutions to adopt systems that can capture and analyze multiple performance indicators. A Student Performance Tracker addresses these challenges by offering a centralized digital platform to collect, store, manage, and analyze student-related data, including examination scores, assignments, attendance records, classroom participation, and behavioral indicators.

By leveraging modern information systems and data analytics, educational institutions can enhance transparency, accountability, and efficiency in academic monitoring. The availability of accurate and up-to-date performance data enables educators to make informed decisions, implement targeted interventions, and support continuous improvement in teaching and learning processes. Thus, the Student Performance Tracker plays a vital role in modernizing academic performance management.

1.2 Problem Statement

Despite the availability of various digital tools in educational institutions, many organizations still lack an integrated system that provides a holistic and unified view of student performance. In many cases, academic marks, attendance records, and assessment data are maintained using separate systems or manual processes. This fragmented approach makes it difficult for educators and administrators to correlate different performance indicators and identify meaningful trends.

As a result, underperforming or at-risk students are often identified at a later stage, when academic difficulties have already intensified. The absence of real-time monitoring and analytical capabilities limits the effectiveness of academic support and intervention strategies. Furthermore, students may not have adequate access to their own performance data, reducing opportunities for self-evaluation and improvement. These challenges highlight the need for a centralized and intelligent Student Performance Tracker that integrates multiple data sources into a single, accessible, and efficient system.

1.3 Objectives of the Project

The primary objective of the Student Performance Tracker is to design and develop an efficient system that enables effective monitoring and analysis of student academic performance. The specific objectives of the project include:

- To design and implement a centralized platform for managing student performance data
- To collect and store academic, attendance, and assessment-related information securely
- To provide analytical insights and performance trends to support academic evaluation
- To assist educators and administrators in identifying underperforming or at-risk students at an early stage
- To enable students to access and track their own academic progress in a transparent manner
- To improve academic planning, reporting, and decision-making processes within educational institutions

These objectives collectively aim to enhance academic performance management and promote data-driven educational practices.

1.4 Scope of the Project

The scope of the Student Performance Tracker is limited to monitoring and analyzing student performance within an educational institution. The system primarily focuses on academic-related parameters such as examination results, internal assessments, attendance records, and assignment submissions. It supports multiple user roles, including students, faculty members, and administrators, each with defined access privileges.

The current scope does not include advanced features such as predictive modeling, artificial intelligence-based recommendations, or integration with external learning management systems. However, the system is designed with scalability in mind, allowing for future enhancements such as predictive analytics, performance forecasting, and personalized learning recommendations to be incorporated at later stages.

1.5 Significance of the Study

The Student Performance Tracker holds significant importance in improving the overall effectiveness of academic monitoring and evaluation processes. By providing timely, accurate, and comprehensive performance data, the system supports early identification of learning gaps and enables proactive intervention strategies. This helps reduce academic failure rates and improves student retention and success.

For administrators, the system offers valuable insights into institutional performance and academic trends, facilitating better planning and policy formulation. Faculty members benefit from simplified performance tracking and reporting, while students gain greater awareness of their academic progress and areas for improvement. Overall, the proposed system contributes to enhanced educational quality, improved learning outcomes, and the adoption of data-driven decision-making in educational management.

CHAPTER 2

LITERATURE SURVEY

2. Literature Survey

2.1 Web-Based Student Performance Monitoring System

AUTHORS: R. K. Sharma , P. Verma & S. Gupta

PUBLISHED ON: 2019, IEEE Conference on Educational Technologies

This paper presents a web-based application designed to monitor student academic performance in educational institutions. The system stores student marks, attendance, and internal assessment data in a centralized database. Faculty members can update records online, while students can view their academic progress. The study highlights improved efficiency and transparency compared to manual systems.

2.2 Online Student Academic Performance Tracking System

AUTHORS: A. Rahman, M. Hossain & T. Ahmed

PUBLISHED ON: 2020, IEEE International Conference on Information Systems

The authors propose an online academic tracking system that allows institutions to manage student performance data through a web interface. The system integrates modules for attendance management, examination results, and progress reports. It enables real-time access to academic records for students and faculty. The paper emphasizes ease of access and reduced administrative workload.

2.3 Web Application for Student Progress Evaluation

AUTHORS: S. Patel, K. Mehta & R. Desai

PUBLISHED ON: 2021, IEEE International Conference on Smart Systems

This research describes a web-based student progress evaluation system developed using standard web technologies. The application allows instructors to upload academic scores and generate performance reports. Students can monitor their progress semester-wise through dashboards. The system improves communication between students and faculty.

2.4 Student Academic Record Management System Using Web Technologies

AUTHORS: N. Kumar & A. Singh

PUBLISHED ON: 2018, IEEE International Conference on Computing

This paper focuses on a web-based academic record management system for maintaining student performance data. The system supports secure login, role-based access, and centralized storage of academic records. It reduces data redundancy and manual errors. The authors conclude that web applications enhance reliability and scalability in academic management.

2.5 Web-Based Performance Tracking System for Higher Education

AUTHORS: P. Johnson, L. Brown & M. Wilson

PUBLISHED ON: 2022, IEEE Access

The study proposes a performance tracking system implemented as a web application for higher education institutions. It provides modules for attendance tracking, internal assessments, and final grades. The system supports report generation for academic review. The authors highlight improved decision-making through structured academic data.

2.6 Online Student Result Analysis and Reporting System

AUTHORS: S. Reddy, V. Rao & K. Prasad

PUBLISHED ON: 2020, IEEE International Conference on Web Applications

This paper introduces a web-based result analysis system that automates the calculation and presentation of student performance. The application allows faculty to upload results and generate reports in graphical and tabular formats. Students can access their results securely. The system enhances speed and accuracy in academic evaluation.

2.7 Web-Based Student Monitoring and Reporting System

AUTHORS: A. Thomas & J. Mathew

PUBLISHED ON: 2019, IEEE International Conference on Software Engineering

The authors present a student monitoring system developed as a web application to track academic and attendance records. The system enables continuous monitoring of student progress throughout the semester. Administrators can generate consolidated reports for analysis. The paper emphasizes improved academic supervision and transparency.

2.8 Academic Performance Management System Using Web Frameworks

AUTHORS: R. Malhotra & S. Bansal

PUBLISHED ON: 2021, IEEE International Conference on Advanced Computing

This research describes the development of a web-based academic performance management system using modern web frameworks. The system supports multi-user access and role-based control. It improves accessibility and data consistency. The authors highlight the flexibility of web applications in educational environments.

2.9 Web-Enabled Student Performance Information System

AUTHORS: D. Chandra & P. Kaur

PUBLISHED ON: 2018, IEEE International Conference on Information Technology

This paper proposes a web-enabled information system for managing student performance records. The system integrates attendance, marks, and academic history into a single platform. Users can retrieve data through a browser-based interface. The system improves data accessibility and administrative efficiency.

2.10 Online Academic Performance Tracking and Reporting Application

AUTHORS: M. Ali, S. Khan & F. Hussain

PUBLISHED ON: 2022, IEEE International Conference on Digital Education

This study presents an online academic performance tracking application that supports continuous assessment and reporting. The system allows students to view their academic status and progress reports. Faculty members can update performance data regularly. The paper concludes that web-based systems enhance student engagement and academic transparency.

CHAPTER 3

PESTAL ANALYSIS

3. Pestal Analysis

PESTAL analysis is a strategic framework used to examine the external factors that influence the design, development, implementation, and adoption of a system. In the context of the Student Performance Tracker, PESTAL analysis helps in understanding how political, economic, social, technological, environmental, and legal factors affect the feasibility and success of a web-based academic performance tracking system.

3.1 Political Factors

Political factors play a significant role in shaping the educational technology landscape. Government policies promoting digital education, e-learning platforms, and smart campus initiatives encourage the adoption of web-based systems in educational institutions. National and regional education boards often mandate standardized reporting and record-keeping practices, which influence system design.

Additionally, policies related to data governance and digital infrastructure development affect how student data is stored and accessed. The Student Performance Tracker must align with government guidelines on educational data management and digital transformation. Supportive political initiatives enhance the adoption of such web applications, while restrictive regulations may require additional compliance measures.

3.2 Economic Factors

Economic factors significantly impact the implementation of web-based systems in educational institutions. Budget constraints, especially in public institutions, often limit the adoption of expensive software solutions. Therefore, the Student Performance Tracker must be cost-effective, requiring minimal infrastructure investment and low maintenance costs.

A web application offers economic advantages by eliminating the need for specialized hardware and allowing access through standard devices such as computers and smartphones. Scalability is also an important economic consideration, as the system should support an increasing number of users without a proportional increase in cost. An affordable and scalable solution improves feasibility and long-term sustainability.

3.3 Social Factors

Social factors influence user acceptance and effective utilization of the Student Performance Tracker. The growing acceptance of digital learning tools among students and educators supports the adoption of web-based academic systems. Increased emphasis on transparency, continuous assessment, and academic accountability further strengthens the need for performance tracking solutions.

However, variations in digital literacy among users must be considered. While students may adapt quickly to online systems, some faculty members may require training and technical support. The system must therefore be user-friendly, intuitive, and accessible to users with different levels of technical expertise to ensure widespread acceptance.

3.4 Technological Factors

Technological advancements play a crucial role in the development of the Student Performance Tracker as a web application. Improvements in web technologies, cloud computing, database management systems, and secure authentication mechanisms enable efficient and reliable system implementation.

The system should be compatible with commonly used web browsers and devices to ensure accessibility. Integration with existing institutional systems, such as student information systems or learning management platforms, is an important technological consideration. Regular updates and maintenance are required to ensure system reliability, performance, and security in a rapidly evolving technological environment.

3.5 Environmental Factors

Environmental considerations support the adoption of web-based academic systems. By reducing dependence on paper-based records, the Student Performance Tracker contributes to environmental sustainability and supports eco-friendly institutional practices. Digital storage of academic data minimizes paper consumption and physical storage requirements.

Remote access to the system reduces the need for physical travel and on-campus administrative processes, indirectly lowering energy consumption. The adoption of web applications aligns with green computing initiatives and promotes sustainable educational practices.

3.6 Legal Factors

Legal factors are critical in the development and deployment of a Student Performance Tracker web application. Educational institutions are required to comply with data protection and privacy regulations governing the storage and processing of student information. Laws related to student data privacy mandate secure handling of academic records.

The system must implement secure authentication, role-based access control, and data encryption to prevent unauthorized access. Compliance with institutional policies and legal standards ensures trust among users and reduces the risk of legal violations. Regular audits and security updates are essential to maintain legal compliance and system integrity.

CHAPTER 4

DESIGN ANYALSIS

4.Design Analysis

Design analysis defines the overall structure and functioning of the Student Performance Tracker web application. This chapter explains the system architecture, design methodology, high-level design, module structure, and constraints considered during development. The design ensures scalability, security, usability, and efficient performance tracking.

4.1 System Architecture

The Student Performance Tracker follows a three-tier modular architecture consisting of the presentation layer, application layer, and data layer. This architecture separates concerns and improves maintainability, scalability, and security.

- The presentation layer handles user interaction through web browsers.
- The application layer processes user requests and implements business logic.
- The data layer manages storage and retrieval of student performance data.

This architecture allows multiple users (students, faculty, administrators) to access the system simultaneously while ensuring secure and controlled data flow.

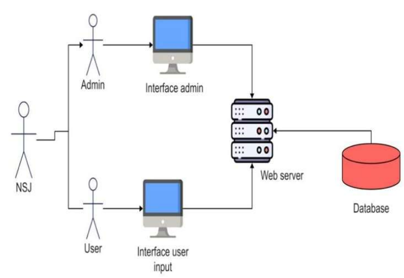

Fig no. 4.1

4.2 Design Methodology

The system design is based on structured analysis and object-oriented design principles. The methodology emphasizes modular development, reusability, and clarity.

Key design techniques used include:

- Use Case Modeling to identify user interactions
- Data Flow Analysis to track data movement within the system
- Modular Decomposition to divide the system into independent components

This approach ensures that the system is easy to understand, modify, and extend in the future.

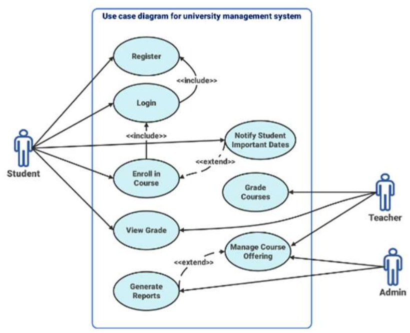

Fig no. 4.2

4.3Module Description

4.3.1 User Interface Module

This module provides the web-based interface for all users. It allows students to view performance details, faculty to update records, and administrators to manage the system.

Functions include:

- User login and authentication
- Dashboard display
- Viewing grades, attendance, and reports

4.3.2 Performance Management Module

This module handles academic data processing and report generation. It calculates attendance percentages, summarizes grades, and generates performance reports.

Functions include:

- Marks and attendance processing
- Performance summary generation
- Report visualization

4.3.3 Database Module

The database module stores all academic records securely. It ensures data consistency, integrity, and quick retrieval.

Stored data includes:

- Student personal information
- Course details
- Marks, attendance, and academic history

4.4 Design Constraints

Several constraints were considered during the design of the Student Performance Tracker:

- **Security Constraints:**
- **Performance Constraints:**

CHAPTER 5

REQUIREMENT ANALYSIS

5.Requirement Analysis

Requirement analysis is a critical phase in the system development process. It involves identifying, analyzing, and documenting the functional and non-functional requirements of the Student Performance Tracker web application. This ensures that the system meets the expectations of all stakeholders and operates efficiently within the institutional environment.

5.1 Overview

The purpose of requirement analysis is to clearly define what the Student Performance Tracker system should do and how it should perform. This process helps in understanding user needs, system constraints, and operational conditions. A well-defined requirement specification reduces ambiguity, minimizes development errors, and ensures the successful implementation of the system.

The requirements are categorized into functional, non-functional, hardware, and software requirements to provide a comprehensive understanding of the system's capabilities and limitations.

5.2 Stakeholder Analysis

The Student Performance Tracker involves multiple stakeholders, each with specific needs and expectations:

- **Students**
- **Faculty Members**
- **Administrators**
- **System Developers**

5.3 Functional Requirements

Functional requirements describe the specific operations that the Student Performance Tracker must perform. These include:

- **User Authentication and Authorization**
- **udent Profile Management**
- **Marks and Attendance Management**
- **Performance Report Generation**
- **Data Visualization and Dashboar**

5.4 Non-Functional Requirements

Non-functional requirements define how the system performs rather than what it does. These include:

- **Security:**
  The system must ensure the confidentiality and integrity of student data. Secure authentication, encrypted data storage, and controlled access mechanisms are mandatory.
- **Performance:**
  The web application should provide fast response times and efficient data retrieval, even when multiple users access the system simultaneously.
- **Usability:**
  The system should have a simple, intuitive, and user-friendly interface. Minimal training should be required for users to operate the system effectively.
- **Scalability:**
  The system should support an increasing number of users and data records without performance degradation. It should allow future enhancements and feature additions.
- **Reliability:**
  The system should be available with minimal downtime and provide consistent performance during regular usage.

5.5 Hardware and Software Requirements

Hardware Requirements

- Server with sufficient processing power and memory
- Client devices such as desktops, laptops, tablets, or smartphones
- Reliable network connectivity

Software Requirements

- **Web server**
- **Database Management System (e.g., MySQL or PostgreSQL)**
- **Server-side development framework**
- **Front-end technologies (HTML, CSS, JavaScript)**
- **Web browser for client access**

These requirements ensure smooth deployment and operation of the web application.

5.6 Feasibility Analysis

The feasibility of the Student Performance Tracker has been evaluated in three aspects:

- **Technical Feasibility:**
  The project is technically feasible as it uses widely available web technologies and database systems. No specialized hardware or advanced tools are required.
- **Economic Feasibility:**
  The system is cost-effective due to the use of open-source tools and minimal infrastructure requirements. Maintenance costs are also relatively low.
- **Operational Feasibility:**
  The system is easy to use and integrates well with existing academic processes. Users can adapt to the system with minimal training.